

MORAWA, Australia

WMO: 944170

Lat: 29.2039S

Lon: 116.0247E

Elev: 892

StdP: 14.23

Time Zone: 8.00 (E08)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.9	39.9	19.4	15.1	75.3	23.2	18.1	76.3	22.6	64.0	20.3	62.3	3.1	170	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.3	105.5	67.1	102.7	66.7	100.0	66.4	74.2	88.0	72.6	88.0	70.8	88.2	9.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.2	118.9	78.5	68.2	106.9	77.6	65.5	97.2	77.2	38.4	87.7	36.8	88.5	35.3	88.1	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.8	19.6	17.8	DB	33.7	110.8	1.6	2.7	32.5	112.7	31.6	114.3	30.8	115.9	29.6	117.8
			WB	31.9	77.5	2.2	3.2	30.3	79.7	29.0	81.6	27.7	83.4	26.1	85.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.6	83.3	82.7	78.2	70.8	62.6	56.6	54.3	55.9	59.4	66.9	74.2	78.8
	DBStd	11.74	6.07	6.03	6.19	5.75	5.08	4.31	3.97	4.33	5.30	6.09	6.68	6.88
	HDD50	15	0	0	0	0	0	2	9	3	1	0	0	0
	HDD65	1230	0	0	1	15	107	251	333	283	185	48	6	1
	CDD50	6792	1031	915	873	624	391	202	142	188	281	525	727	893
	CDD65	2531	566	496	409	188	33	1	0	2	16	108	283	429
	CDH74	32356	7723	6505	4730	1812	357	22	5	66	331	1521	3616	5668
	CDH80	18152	4853	3896	2537	733	75	1	0	8	99	637	1965	3348
(14) Wind	WSAvg	8.7	11.1	10.4	10.0	8.1	7.0	6.9	6.7	6.9	8.0	9.0	10.1	10.7
(15) Precipitation	PrecAvg	13.0	0.6	0.7	0.8	0.9	2.0	2.3	2.0	1.4	0.9	0.6	0.4	0.4
	PrecMax	20.5	3.7	4.3	8.7	4.2	6.6	6.9	5.0	4.4	2.6	3.2	1.8	1.4
	PrecMin	6.8	0.0	0.0	0.0	0.0	0.1	0.3	0.2	0.3	0.2	0.0	0.0	0.0
	PrecStd	3.3	0.8	0.9	1.5	1.0	1.6	1.2	1.1	0.7	0.6	0.6	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	110.1	107.5	103.8	96.1	86.3	76.8	73.7	80.0	88.3	96.7	103.6	107.4
		MCWB	66.8	67.6	65.7	63.8	60.2	56.3	55.8	56.4	59.4	61.6	65.0	66.3
	2%	DB	106.3	103.5	99.9	91.8	81.6	73.2	69.6	75.0	82.8	91.8	99.3	103.7
		MCWB	67.7	67.6	66.2	62.6	58.6	54.6	54.8	55.6	57.7	61.3	63.1	65.8
	5%	DB	103.0	100.8	96.5	87.9	77.8	69.9	66.6	70.7	77.5	87.0	95.5	99.9
		MCWB	67.1	67.6	66.3	62.4	57.9	54.9	54.6	55.1	56.7	59.8	62.1	65.5
	10%	DB	99.7	97.7	92.9	84.1	74.3	66.6	63.8	67.1	72.7	82.6	91.4	96.0
		MCWB	67.1	67.9	65.5	61.9	57.4	54.6	54.3	54.8	56.1	58.9	61.3	64.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.6	76.3	73.7	69.2	66.7	63.1	62.5	62.3	63.1	65.8	68.9	72.8
		MCDB	85.0	88.4	83.1	81.3	72.7	66.3	64.7	65.7	74.7	82.8	89.4	90.3
	2%	WB	74.4	74.4	71.5	67.5	64.0	60.9	59.3	60.2	61.2	63.6	66.3	70.0
		MCDB	89.3	87.4	83.5	78.7	71.5	66.2	62.7	66.1	72.8	81.1	89.1	92.0
	5%	WB	72.7	72.7	69.9	66.0	62.0	59.0	57.7	58.5	59.6	62.1	64.8	68.2
		MCDB	89.3	87.9	86.0	79.0	70.1	64.0	61.8	65.1	70.7	79.2	85.5	90.3
	10%	WB	70.9	70.8	68.2	64.4	60.3	57.4	56.2	56.8	58.0	60.8	63.5	66.5
		MCDB	88.5	89.2	84.9	77.1	69.3	62.6	61.1	63.7	68.5	77.1	83.6	89.7
(35) Mean Daily Temperature Range	5% DB	MDBR	29.3	26.8	25.3	24.4	23.0	20.6	20.4	22.8	25.7	28.4	28.9	29.1
		MCDBR	33.6	30.3	29.9	29.5	28.3	25.4	25.0	29.8	34.1	34.9	34.5	33.7
		MCWBR	10.1	8.8	8.8	10.1	11.7	12.2	13.1	13.6	13.3	11.4	10.6	10.2
	5% WB	MCDBR	24.6	23.9	23.7	22.6	19.3	17.0	17.2	20.8	25.9	28.7	29.9	29.4
		MCWBR	8.2	8.3	8.3	8.8	10.0	10.2	11.4	11.5	11.5	10.1	9.6	10.2
(40) Clear-Sky Solar Irradiance	taub		0.335	0.341	0.323	0.310	0.292	0.283	0.282	0.287	0.306	0.312	0.318	0.325
	taud		2.553	2.554	2.584	2.599	2.617	2.638	2.629	2.607	2.523	2.523	2.532	2.535
	Ebn at Noon		319	312	308	297	288	284	289	300	308	317	323	323
	Edh at Noon		35	34	31	27	24	23	23	27	32	34	35	35
(44) All-Sky Solar Radiation	RadAvg		2594	2348	2005	1563	1219	1027	1090	1385	1805	2236	2535	2687
	RadStd		150	122	128	75	75	68	61	45	81	94	94	128

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page